

Applications of Trigonometry

Chapter 9 - Heights & Distances, Hinglish Notes + Solutions

Ye Chapter 9 (Some Applications of Trigonometry) ka complete guide hai. Isme trigonometry ka use height aur distance nikalne me hota hai. Pehle red-page ke core basics, fir main content - line of sight, angle of elevation/depression aur problems solve karna. End me solved examples aur 20 practice questions ke step-by-step solutions hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

PART A - Core Basics (Miss Mat Karna)

1. Trig Ratios (Chapter 8 Recap)

EXAM * Exam me CONSTANTLY aata hai

- $\sin = \text{Perpendicular} / \text{Hypotenuse}$
- $\cos = \text{Base} / \text{Hypotenuse}$
- $\tan = \text{Perpendicular} / \text{Base}$

2. Special Angle Values (Zaroori)

EXAM * Exam me CONSTANTLY aata hai

- $\tan 30 = 1/\sqrt{3}$, $\tan 45 = 1$, $\tan 60 = \sqrt{3}$.
- $\sin 30 = 1/2$, $\sin 45 = 1/\sqrt{2}$, $\sin 60 = \sqrt{3}/2$.
- $\cos 30 = \sqrt{3}/2$, $\cos 45 = 1/\sqrt{2}$, $\cos 60 = 1/2$.

3. Square Root Values

- $\sqrt{3} = 1.732$, $\sqrt{2} = 1.414$ (decimal answer ke liye).

4. Horizontal aur Vertical

EXAM * Exam me CONSTANTLY aata hai

- Horizontal = leti line (zameen ke parallel).
- Vertical = khadi line (tower/building ki height).

5. Right Triangle Banta Hai

EXAM * Exam me CONSTANTLY aata hai

- Tower (vertical) + zameen (horizontal) + line of sight milke ek right-angled triangle banate hain.

6. Alternate Angles (Parallel Lines)

EXAM * Exam me CONSTANTLY aata hai

- Jab do horizontal lines parallel ho, to angle of elevation = angle of depression (alternate angles).

7. Solve for Unknown

EXAM * Exam me CONSTANTLY aata hai

- $\tan(\text{angle}) = \text{opposite} / \text{adjacent}$.
- Ek side pata ho to doosri multiply/divide karke nikaal lo.

PART B - Chapter 9 ka Main Content

8. Line of Sight

EXAM * Exam me CONSTANTLY aata hai

Definition: Observer ki aankh se object tak ki seedhi (straight) line ko line of sight kehte hain.

9. Angle of Elevation

EXAM * Exam me CONSTANTLY aata hai

Definition: Jab object UPAR ho, to horizontal line se line of sight tak UPAR ki taraf bana angle = angle of elevation.

- Example: zameen par khade ho kar tower ki choti dekhna.

10. Angle of Depression

EXAM * Exam me CONSTANTLY aata hai

Definition: Jab object NEECHE ho, to horizontal line se line of sight tak NEECHE ki taraf bana angle = angle of depression.

- Example: building ki chhat se neeche khadi car dekhna.
- Yaad: elevation = upar dekho, depression = neeche dekho.

11. Problem Solve Karne ke Steps

EXAM * Exam me CONSTANTLY aata hai

- Step 1: Diagram banao - vertical (height), horizontal (distance), angle mark karo.
- Step 2: Right triangle pehchaano; kaunsi side pata hai aur kya nikaalna hai.
- Step 3: Sahi ratio chuno ($\tan$, $\sin$ ya $\cos$) angle aur sides ke hisaab se.
- Step 4: Equation banao aur unknown solve karo.

12. Kaunsa Ratio Kab

Height aur Base (horizontal distance) involved -> $\tan$ use karo.

Height aur Hypotenuse (line of sight) involved -> $\sin$ use karo.

Base aur Hypotenuse involved -> $\cos$ use karo.

Quick Revision

Elevation = upar dekhne wala angle; Depression = neeche dekhne wala.

$\tan(\text{angle}) = \text{height} / \text{base}$ (sabse zyada use hota hai).

Special: $\tan 30 = 1/\sqrt{3}$, $\tan 45 = 1$, $\tan 60 = \sqrt{3}$.

Elevation = depression (alternate angles, parallel ground).

Hamesha pehle diagram banao!

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain.

Example 1: Tower se 30m door se choti ka elevation 45 hai. Tower ki height?

- Step: $\tan 45 = \text{height} / \text{base}$.
- Step: $1 = h / 30$.
- Step: $h = 30 \times 1$.

Answer: Height = 30 m

Example 2: 20m door se elevation 60. Tower ki height?

- Step: $\tan 60 = h / 20 \rightarrow \sqrt{3} = h / 20$.
- Step: $h = 20 \sqrt{3}$.
- Step: $= 20 \times 1.732$.

Answer: Height = $20 \sqrt{3} \sim 34.64$ m

Example 3: Tower height 30m, base se distance $30 \sqrt{3}$ m. Elevation angle?

- Step: $\tan(\text{angle}) = 30 / (30 \sqrt{3}) = 1/\sqrt{3}$.
- Step: $\tan 30 = 1/\sqrt{3}$.

Answer: Angle = 30 degree

Example 4: Pole height 10m. Elevation 30 ho to distance?

- Step: $\tan 30 = 10 / d \rightarrow 1/\sqrt{3} = 10/d$.
- Step: $d = 10 \sqrt{3}$.

Answer: Distance = $10 \sqrt{3} \sim 17.32$ m

Example 5: 100m building ki chhat se car ka depression 45. Car ki doori?

- Step: Depression 45 = elevation 45 (alternate).
- Step: $\tan 45 = 100 / d \rightarrow 1 = 100/d$.

Answer: Distance = 100 m

Example 6: Ladder 10m, wall se elevation 60. Wall par height?

- Step: $\sin 60 = \text{height} / 10$ (ladder = hypotenuse).
- Step: $\sqrt{3}/2 = h/10 \rightarrow h = 5 \sqrt{3}$.

Answer: Height = $5 \sqrt{3} \sim 8.66$ m

Example 7: Kite ki dori 50m, elevation 30. Kite ki height?

- Step: $\sin 30 = h / 50$ (dori = hypotenuse).
- Step: $1/2 = h/50 \rightarrow h = 25$.

Answer: Height = 25 m

Example 8: Elevation 45 par height = distance hota hai. Sahi ya galat?

- Step: $\tan 45 = 1 \rightarrow \text{height}/\text{base} = 1$.
- Step: Isliye height = base.

Answer: Sahi

Example 9: Ladder 10m, base se 5m door. Wall se angle (cos use)?

- Step: $\cos(\text{angle}) = \text{base} / \text{hyp} = 5/10 = 1/2$.
- Step: $\cos 60 = 1/2$.

Answer: Angle = 60 degree

Example 10: Tower height 15m, elevation 30. Observer ki doori?

- Step: $\tan 30 = 15 / d \rightarrow 1/\sqrt{3} = 15/d$.
- Step: $d = 15 \sqrt{3}$.

Answer:Distance = $15\sqrt{3} \sim 25.98$ m

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

- Q1. Line of sight kya hota hai?
- Q2. Angle of elevation kya hota hai?
- Q3. Angle of depression kya hota hai?
- Q4. Height aur base diya ho to kaunsa ratio use karte hain?
- Q5. $\tan 45$ ka value kya hai?
- Q6. $\tan 60$ ka value kya hai?
- Q7. $\tan 30$ ka value kya hai?
- Q8. Elevation aur depression me kya relation hota hai (parallel ground)?
- Q9. 30m door se elevation 45 $\rightarrow$ tower height?
- Q10. Tower height 10m, elevation 45 $\rightarrow$ distance?
- Q11. Height aur hypotenuse diya ho to kaunsa ratio?
- Q12. Base aur hypotenuse diya ho to kaunsa ratio?
- Q13. $\sqrt{3}$ ka approx value kya hai?
- Q14. Problem solve karne ka pehla step kya hona chahiye?
- Q15. Elevation 60, base 10m $\rightarrow$ height?
- Q16. Ladder 10m hypotenuse, $\sin 30 \rightarrow$ wall par height?
- Q17. Tower height = distance kab hota hai?
- Q18. Depression angle kahan se naapte hain (upar ya neeche)?
- Q19. $\tan(\text{angle})$ kis ke barabar hota hai (opposite/adjacent)?
- Q20. Vertical aur horizontal ka matlab batao.

Step-by-Step Solutions

- Q1. Line of sight?
 - Step: Aankh se object tak seedhi line.**Answer:**Observer se object tak straight line
- Q2. Angle of elevation?
 - Step: Object upar; horizontal se upar ka angle.**Answer:**Upar dekhne wala angle
- Q3. Angle of depression?
 - Step: Object neeche; horizontal se neeche ka angle.**Answer:**Neeche dekhne wala angle
- Q4. Height + base $\rightarrow$ kaunsa ratio?
 - Step: $\tan = \text{opp/adj}$.**Answer:** $\tan$
- Q5. $\tan 45$?
 - Step: Table se.

Answer:1

Q6. tan 60?

- Step: Table se.

Answer:sqrt3

Q7. tan 30?

- Step: Table se.

Answer:1/sqrt3

Q8. Elevation vs depression?

- Step: Alternate angles (parallel ground).

Answer:Dono barabar hote hain

Q9. 30m door, elevation 45 -> height?

- Step: $\tan 45 = h/30 \rightarrow 1 = h/30$.

Answer:30 m

Q10. Height 10m, elevation 45 -> distance?

- Step: $\tan 45 = 10/d \rightarrow 1 = 10/d$.

Answer:10 m

Q11. Height + hypotenuse -> ratio?

- Step: $\sin = \text{opp}/\text{hyp}$.

Answer:sin

Q12. Base + hypotenuse -> ratio?

- Step: $\cos = \text{adj}/\text{hyp}$.

Answer:cos

Q13. sqrt3 approx?

- Step: Yaad value.

Answer:1.732

Q14. Pehla step?

- Step: Situation samajhne ke liye.

Answer:Diagram banao

Q15. Elevation 60, base 10 -> height?

- Step: $\tan 60 = h/10 \rightarrow \sqrt{3} = h/10$.

Answer:10 sqrt3 (~17.32 m)

Q16. Ladder 10m, sin30 -> height?

- Step: $\sin 30 = h/10 \rightarrow 1/2 = h/10$.

Answer:5 m

Q17. Height = distance kab?

- Step: $\tan 45 = 1 \rightarrow h = \text{base}$.

Answer:Jab angle 45 ho

Q18. Depression kahan se?

- Step: Upar ki horizontal se neeche.

Answer:Upar se (neeche ki taraf)

Q19. tan(angle) = ?

- Step: Right triangle ratio.

Answer:opposite / adjacent

Q20. Vertical aur horizontal?

- Step: Khadi aur leti line.

Answer:Vertical = khadi, Horizontal = leti